

1. Upload files into galaxy
2. Select all (remove the dataset), then auto build list, and rename the file
3. dada2: plotQualityProfile (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: The dataset collection
 - b. Parameter: Default
 - c. Download: The reverse and forward plots
4. dada2: filterAndTrim (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: The dataset collection
 - b. Parameter: Default (all the data was above 30)
 - c. Download: N/A
5. Unzip collection (Galaxy Version 1.0.0)
 - a. Upload: the filter and trim
 - b. Parameter: Default
 - c. Download: N/A
6. dada2: learnErrors (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: Forward Dataset collection
 - b. Parameter: Default
 - c. Download: The forward error plot
 - d. Edit it and add “forward” to the name
7. dada2: learnErrors (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: Reverse Dataset collection
 - b. Parameter: Default
 - c. Download: The reverse error plot
 - d. Edit it and add “reverse” to the name
8. dada2: dada (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: Forward collection and forward learnErrors
 - b. Parameter: Process samples in batches → yes
 - c. Download: N/A
 - d. Edit it and add “forward” to the name
9. dada2: dada (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: Reverse collection and reverse learnErrors
 - b. Parameter: Process samples in batches → yes
 - c. Download: N/A
 - d. Edit it and add “reverse” to the name
10. dada2: mergePairs (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: Forward dada results and forward dataset collection; reverse dada results and reverse dataset collection
 - b. Parameter: Default
 - c. Download: N/A
11. dada2: makeSequenceTable (Galaxy Version 1.38.0+galaxy0)

- a. Upload: mergePairs output
 - b. Parameter: Default
 - c. Download: Both outputs (the table and plot)
- 12. dada2: removeBimeraDenovo (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: makeSequenceTable output
 - b. Parameter: Default
 - c. Download: The output
- 13. dada2: assignTaxonomy and addSpecies (Galaxy Version 1.38.0+galaxy0)
 - a. Upload: removeBimeraDenovo output
 - b. Parameter: Select reference data set → silva version 138; Add genus-species binomials to the taxonomic table → yes; Select reference data set → silva version 138
 - c. Download: The output
- 14. Create phyloseq object (Galaxy Version 1.54.0+galaxy0)
 - a. Upload: removeBimeraDenovo output, assignTaxonomy output, metadata
 - b. Parameter: Default
 - c. Download: N/A
- 15. Phyloseq: Bar Chart (Galaxy Version 1.54.0+galaxy0)
 - a. Upload: Create phyloseq output
 - b. Parameter: Fill variable → Phylum; Facet by variable → Source; Keep 'Others' → yes; Keep 'Not Assigned' → yes; Normalize → yes
 - c. Download: The output
- 16. Go to interactive tools → Type in phyloseq → Upload your phyloseq object → click the three lines → click alpha diversity → unclick Chao1 and click Observed and Simpson → add to the graph as you like → download the final plot
- 17. In the same phyloseq tool → Ordination → change method from DCA to NMDS → add to graph as you like → download the final plot